

# AI ASSISTANT USAGE POLICY

## Philosophy

### **AI tools are learning aids, not replacement for learning.**

You are **encouraged** to use ChatGPT, Claude, GitHub Copilot, and other AI assistants – but you must use them to **learn**, not to avoid learning.

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## ALLOWED (and Encouraged!)

### 1. Use AI to Learn and Understand

#### **Examples:**

- "Explain how YOLO object detection works"
- "What's the difference between Faster R-CNN and YOLO?"
- "How does transfer learning work?"
- "Why do we use non-max suppression?"
- "Explain this error message: [paste error]"

#### **Why this is good:**

- You're building understanding
  - You can ask follow-up questions
  - You learn at your own pace
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### 2. Use AI for Skills You Don't Have

#### **Examples:**

- "Generate JavaScript code to create an interactive visualization" (when you only know Python)
- "Write a Dockerfile for my Python CV application" (when you've never used Docker)
- "Create a bash script to automate data download" (when you don't know bash)
- "Generate CSS styling for my demo webpage" (when you don't know web development)

#### **Why this is good:**

- These aren't the learning objectives of this course
  - You can learn these skills later if needed
  - Focuses your time on CV concepts
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### 3. Use AI to Debug

**Examples:**

- "Why am I getting this error? [paste error and code]"
- "My YOLO model isn't training, help me debug"
- "This code works but runs slowly, how to optimize?"
- "Why are my bounding boxes not showing up?"

**Why this is good:**

- Debugging is frustrating and time-consuming
  - AI can spot issues you might miss
  - You learn from the solution
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### 4. Use AI to Improve Code

**Examples:**

- "How can I make this code more efficient?"
- "Is there a better way to structure this?"
- "Add error handling to this function"
- "Make this code more readable"

**Why this is good:**

- You learn best practices
  - Your code quality improves
  - Professional development skill
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### 5. Use AI for Brainstorming

**Examples:**

- "What are interesting CV project ideas for retail?"
- "Where can I find datasets for plant disease detection?"
- "What metrics should I use to evaluate object detection?"
- "How can I improve my model's accuracy?"

**Why this is good:**

- Helps overcome creative blocks
  - Exposes you to options
  - Final decision is still yours
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# NOT ALLOWED

## 1. Don't Let AI Do Your Project

### Examples:

- "Build me a complete object detection system for inventory"
- "Write all the code for my CV project from start to finish"
- "Generate my entire project proposal"
- "Create my presentation slides about [my project]"

### Why this is bad:

- You won't learn anything
  - You won't be able to explain your code
  - You'll fail Q&A during presentation
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## 2. Don't Copy-Paste Without Understanding

### Bad Workflow:

- Ask AI for code
- Copy entire code block
- Paste into your project
- Run it without reading
- Submit without understanding

### Good Workflow:

- Ask AI for code
  - Read through it carefully
  - Ask AI to explain confusing parts
  - Modify it for your needs
  - Add comments explaining what it does
  - Test it thoroughly
  - Be ready to explain every line
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## 3. Don't Use AI for Core CV Concepts

### Examples:

- "Implement YOLO from scratch for me"
- "Write the training loop for my CNN"
- "Build a custom loss function for my model"

## Why this is bad:

- These ARE the learning objectives
- You need to understand these deeply
- Okay to use pre-built libraries (PyTorch, Ultralytics)
- Not okay to have AI write your CV algorithms



## REQUIRED: AI Usage Log

You **must** document your AI usage in your GitHub repository.

## Location:

```
project-name/  
└─ docs/  
    └─ AI_usage_log.md
```

## Template:

```
# AI Usage Log  
  
## 1. Understanding YOLO Architecture  
- **Date**: October 20, 2025  
- **Tool Used**: ChatGPT  
- **Question Asked**: "Explain how YOLOv8 works and why it's good for real-time detection"  
- **What I Learned**:  
  [2-3 sentences explaining what you understood]  
- **How I Applied It**:  
  [How you used this knowledge in your project]  
  
## 2. Debugging Data Loading Error  
- **Date**: October 25, 2025  
- **Tool Used**: GitHub Copilot  
- **Problem**: Getting dimension mismatch error when loading images  
- **Solution**: [What the AI suggested and why it worked]  
- **What I Learned**: [What you now understand about image preprocessing]
```

## Minimum Required Entries: 5-10

Each entry should show:

1. What you asked/problem you had
2. What the AI suggested

3. What you learned
  4. How you applied it
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## VERIFICATION: How We Know You Learned

### During Proposal (Midterm):

- Can you explain your technical approach?
- Do you understand why you chose that model?
- Can you answer questions about your plan?

### During Presentation (Final):

- Can you run a live demo?
- Can you explain how your code works?
- Can you answer technical questions?
- Can you debug issues on the spot?

### In Your Code:

- Do you have meaningful comments?
  - Are variable names sensible?
  - Does the code structure make sense?
  - Is there evidence of iteration/debugging?
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## AI TOOLS WE RECOMMEND

### Free Options:

#### 1. **ChatGPT** (Free tier)

- Best for: Explanations, brainstorming
- chatgpt.com

#### 2. **Claude** (Free tier)

- Best for: Code review, detailed explanations
- claude.ai

#### 3. **GitHub Copilot** (Free for students)

- Best for: Code completion, boilerplate
- Get student pack: education.github.com

#### 4. **Perplexity AI** (Free tier)

- Best for: Research, finding resources
- perplexity.ai

## Paid Options (Optional):

- ChatGPT Plus (20/month) – *Faster, better model* – ClaudePro(20/month) - Higher usage limits
- GitHub Copilot (\$10/month) - If not student

**Note:** Free tiers are sufficient for this course!

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## BEST PRACTICES

### 1. Start with Understanding

Good: "Explain concept X" → Learn → Apply  
Bad: "Write code for X" → Copy → Submit

### 2. Iterate and Refine

Good Conversation:

You: "How do I load a dataset for YOLO?"  
AI: [Gives example code]  
You: "What if my images are in different folders?"  
AI: [Adjusts solution]  
You: "Can you explain what glob() does?"  
AI: [Explains]

Bad Conversation:

You: "Write complete data loading code"  
AI: [Gives code]  
You: [Copies and pastes]

### 3. Test and Verify

- Always test AI-generated code
- Verify AI explanations with documentation
- Cross-check facts from multiple sources
- Don't trust AI blindly

### 4. Give Context

Good Prompt:

"I'm building an object detection system using YOLOv8 in PyTorch. I have 500 images of products on shelves. How should I structure my dataset folders and create the data.yaml file?"

Bad Prompt:

"How to organize dataset?"

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## ACADEMIC INTEGRITY

### This is Cheating:

- Having AI write your entire project
- Copying code you don't understand
- Not documenting AI usage
- Lying about what you did vs what AI did
- Submitting AI work as your own

### This is Not Cheating:

- Using AI to learn concepts
- Getting debugging help from AI
- Having AI generate non-CV code (HTML, bash scripts)
- Improving your code with AI suggestions
- Using AI as a study buddy

**When in doubt, ASK THE INSTRUCTOR!**

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## FAQ

**Q: Can I use AI to write my README?**

A: AI can help structure it, but the content should be yours. It's about YOUR project.

**Q: What if AI gives me wrong information?**

A: Always verify! Check documentation, test code, cross-reference multiple sources.

**Q: Do I need to cite every time I use AI?**

A: No, just major interactions in your AI usage log. Don't need to log every small question.

**Q: Can I use AI during the presentation?**

A: No! You should know your project well enough to present without AI help.

**Q: What if my teammate used AI but didn't document it?**

A: Each team member is responsible for their AI usage documentation. Document your own use clearly.

**Q: Is using pre-trained models from Hugging Face considered AI assistance?**

A: No! Using libraries and pre-trained models is expected and encouraged. AI assistance refers to ChatGPT/Claude/Copilot, not model libraries.



## CHECKLIST

Before submitting, verify:

- ☐ I created `docs/AI_usage_log.md`
- ☐ I documented 5-10 major AI interactions
- ☐ Each entry has: question, answer, learning, application
- ☐ I can explain all code in my project
- ☐ I understand the CV concepts I implemented
- ☐ I tested all AI-generated code
- ☐ I'm ready to answer questions about my code
- ☐ I added comments explaining AI-assisted sections



## REMEMBER

**AI is a tool, not a shortcut.**

The goal is to **learn computer vision**, not to complete a project as fast as possible.

Use AI to:

- ☒ Learn faster
- ☒ Debug smarter
- ☒ Build better

Don't use AI to:

- ☒ Avoid learning
- ☒ Skip understanding
- ☒ Take shortcuts

**You'll present this project. Make sure you understand it!** 

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**Questions about AI usage? Ask the instructor!**



